

OCR Further Mathematics
Mechanics - Dimensional Analysis
Question Paper
AS and A Level
Further Mathematics A - H235, H245



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The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.

- 1(a) A particle P of mass m moves on the x -axis under the action of a force F directed along the axis. When the displacement of P from the origin is x its velocity is v .

By using the fact that the dimensions of the derivative $\frac{dv}{dx}$ are the same as those of $\frac{v}{x}$, verify that the equation $F = mv \frac{dv}{dx}$ is dimensionally consistent. [2]

- (b) It is given that $v = km^{-\frac{1}{2}} \sqrt{a^2 - x^2}$ where a and k are constants. Explain why $[a]$ must be the same as $[x]$. [1]

- (c) Deduce the dimensions of k . [2]

- (d) Find an expression for F in terms of x and k . [3]

- 2(a) The escape speed of an unpowered object is the minimum speed at which it must be projected to escape the gravitational influence of the Earth if it is projected vertically upwards from the Earth's surface. A formula for the

escape speed U of an unpowered object of mass m is $U = \sqrt{\frac{2Gm}{r}}$ where r is the radius of the Earth and G is a constant.

Show that the dimensions of G are $M^{-1} L^3 T^{-2}$. [3]

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- (b) A rocket is a powered object. A rocket is launched with a given launch speed and is then powered by engines which apply a constant force for a period of time after the launch.

A student wishes to apply the formula given above to a rocket launch. They wish to model the minimum launch speed required for a rocket to escape the Earth's gravitational influence.

They realise that the given formula is for unpowered objects and so they include an extra term in the formula to

obtain $V = \sqrt{\frac{2Gm}{r}} - kP^\alpha W^\beta t^\gamma$.

In their modified formula, G and r are the same as before. The other variables are defined as follows.

- ♦ V is the required minimum launch speed of the rocket
- ♦ k , α , β and γ are dimensionless constants
- ♦ P is the power developed by the engines of the rocket
- ♦ m is the initial mass of the rocket
- ♦ W is the initial weight of the rocket
- ♦ t is the total time for which the engines of the rocket operate

Use dimensional analysis to determine the values of α , β and γ . [6]

- (c) By considering the value of γ found in the above part explain the relationship between t and V . [1]

- 3(a) The resistive force, F , on a sphere falling through a viscous fluid is thought to depend on the radius of the sphere, r , the velocity of the sphere, v , and the viscosity of the fluid, η . You are given that η is measured in $\text{N m}^{-2} \text{s}$.

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By considering its units, find the dimensions of viscosity. [2]

- (b) A model of the resistive force suggests the following relationship: $F = 6\pi h^\alpha r^\beta v^\gamma$.

Explain whether or not it is possible to use dimensional analysis to verify that the constant 6π is correct.

[1]

- (c) Use dimensional analysis to find the values of α , β and γ .

[4]

- (d) A sphere of radius r and mass m falls vertically from rest through the fluid. After a time t its velocity is v .

By setting up and solving a differential equation, show that $e^{-kt} = \frac{g - kv}{g}$ where $k = \frac{6\pi\eta r}{m}$.

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- (e) As the time increases, the velocity of the sphere tends towards a limit called the terminal velocity.

Find, in terms of g and k , the terminal velocity of the sphere.

[1]

- (f) In a sequence of experiments the sphere is allowed to fall through fluids of different viscosity, ranging from small to very large, with all other conditions being constant. The terminal velocity of the sphere through each fluid is measured.

Describe how, according to the model, the terminal velocity of the sphere changes as the viscosity of the fluid through which it falls increases.

[2]



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- 4(a) A particle of mass m moves in a straight line with constant acceleration a . Its initial and final velocities are u and v respectively and its final displacement from its starting position is s . In order to model the motion of the particle it is suggested that the velocity is given by the equation

$$v^2 = pu^\alpha + qa^\beta s^\gamma$$

where p and q are dimensionless constants.

Explain why α must equal 2 for the equation to be dimensionally consistent.

[2]



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(b) By using dimensional analysis, determine the values of β and γ .

[4]

(c) By considering the case where $s = 0$, determine the value of p .

[1]

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(d)

By multiplying both sides of the equation by $\frac{1}{2}m$, and using the numerical values of α , β and γ , determine the value of q .

[2]

- 5(a) A solenoid is a device formed by winding a wire tightly around a hollow cylinder so that the wire forms (approximately) circular loops along the cylinder (see diagram).



When the wire carries an electrical current a magnetic field is created inside the solenoid which can cause a particle which is moving inside the solenoid to accelerate.

A student is carrying out experiments on particles moving inside solenoids. His professor suggests that, for a particle of mass m moving with speed v inside a solenoid of length h , the acceleration a of the particle can be modelled by a relationship of the form $a = km^\alpha v^\beta h^\gamma$, where k is a constant. The professor tells the student that $[k] = \text{MLT}^{-1}$.

Use dimensional analysis to find α , β and γ .

[6]

- (b) The mass of an electron is $9.11 \times 10^{-31}\text{kg}$ and the mass of a proton is $1.67 \times 10^{-27}\text{kg}$.

For an electron and a proton moving inside the same solenoid with the same speed, use the model to find the ratio of the acceleration of the electron to the acceleration of the proton.

[3]



- (c) The professor tells the student that α also depends on the number of turns or loops of wire, N , that the solenoid has.

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Explain why dimensional analysis cannot be used to determine the dependence of α on N .

[1]

6(a) A student is studying the speed of sound, u , in a gas under different conditions.

He assumes that u depends on the pressure, p , of the gas, the density, ρ , of the gas and the wavelength, λ , of the sound in the relationship $u = kp^\alpha \rho^\beta \lambda^\gamma$, where k is a dimensionless constant. (The wavelength of a sound is the distance between successive peaks in the sound wave.)

Use the fact that density is mass per unit volume to find $[\rho]$.

[1]

(b) Given that the units of p are Nm^{-2} , determine the values of α , β and γ .

[7]



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- (c) Comment on what the value of γ means about how fast sounds of different wavelengths travel through the gas. [1]

- (d) The student carries out two experiments, *A* and *B*, to measure u . Only the density of the gas varies between the experiments, all other conditions being unchanged. He finds that the value of u in experiment *B* is double the value in experiment *A*.

By what factor has the density of the gas in experiment *A* been multiplied to give the density of the gas in experiment *B*? [2]



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- 7 This question is about modelling the relation between the pressure, P , volume, V , and temperature, θ , of a fixed amount of gas in a container whose volume can be varied.

The amount of gas is measured in moles; 1 mole is a dimensionless constant representing a fixed number of molecules of gas. Gas temperatures are measured on the Kelvin scale; the unit for temperature is denoted by K. You may assume that temperature is a dimensionless quantity.

A gas in a container will always exert an outwards force on the walls of the container. The pressure of the gas is defined to be the magnitude of this force per unit area of the walls, with P always positive.

An initial model of the relation is given by $P^\alpha V^\beta = nR\theta$, where n is the number of moles of gas present and R is a quantity called the Universal Gas Constant. The value of R , correct to 3 significant figures, is 8.31 J K^{-1} .

- (a) Show that $[P] = \text{ML}^{-1} \text{T}^{-2}$ and $[R] = \text{ML}^2 \text{T}^{-2}$.

[4]



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(b) Hence show that $\alpha = 1$ and $\beta = 1$

[3]

5 moles of gas are present in the container which initially has volume 0.03m^3 and which is maintained at a temperature of 300 K.

(c) Find the pressure of the gas, as predicted by the model.

[2]

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An improved model of the relation is given by $\left(P + \frac{an^2}{V^2}\right)(V - nb) = nR\theta$, where a and b are constants.

(d) Determine the dimensions of b and a .

[3]

The values of a and b (in appropriate units) are measured as being 0.14 and 3.2×10^{-5} respectively.

- (e) Find the pressure of the gas as predicted by the improved model.

[2]

Suppose that the volume of the container is now reduced to $1.5 \times 10^{-4} \text{ m}^3$ while keeping the temperature at 300 K.

- (f) By considering the value of the pressure of the gas as predicted by the improved model, comment on the validity of this model in this situation.

[3]

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- 8 One end of a non-uniform rod is freely hinged to a fixed point so that the rod can rotate about the point. When the rod rotates with angular velocity ω it can be shown that the kinetic energy E of the rod is given

by $E = \frac{1}{2}I\omega^2$, where I is a quantity called the moment of inertia of the rod.

- (a) Deduce the dimensions of I .

[3]



- (b) Given that the rod has mass m and length r , suggest an expression for I , explaining any additional symbols that you use.

[3]

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A student notices that the formula $E = \frac{1}{2}I\omega^2$ looks similar to the formula $E = \frac{1}{2}mv^2$ for the kinetic energy of a particle, with angular velocity for the rod corresponding to velocity for the particle, and moment of inertia

corresponding to mass. Assuming a similar correspondence between angular acceleration (i.e. $\frac{d\omega}{dt}$) and acceleration, the student thinks that an equation for angular motion of the rod corresponding to Newton's second law for the particle should be $F = I\alpha$, where F is the force applied to the rod and α is the resulting angular acceleration.

(c) Use dimensional analysis to show that the student's suggestion is incorrect.

[2]



(d) State the dimensions of a quantity x for which the equation $Fx = I\alpha$ would be dimensionally consistent.

[1]

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(e) Explain why the fact that the equation in part (d) is dimensionally consistent does not necessarily mean that it is correct.

[1]

- 9 A particle moves in a straight line with constant acceleration. Its initial and final velocities are u and v respectively and at time t its displacement from its starting position is s . An equation connecting these quantities is $s = k(u^\alpha + v^\beta)t^\gamma$, where k is a dimensionless constant.

(a) Use dimensional analysis to find the values of α , β and γ .

[6]



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(b) By considering the case where the acceleration is zero, determine the value of k .

[2]

- 10 A simple pendulum consists of a small sphere of mass m connected to one end of a light rod of length h . The

other end of the rod is freely hinged at a fixed point. When the sphere is pulled a short distance to one side and released from rest the pendulum performs oscillations. The time taken to perform one complete oscillation is called the period and is denoted by P .

- (a) Assuming that $P = km^\alpha h^\beta g^\gamma$, where g is the acceleration due to gravity and k is a dimensionless constant, find the values of α , β and γ . [5]



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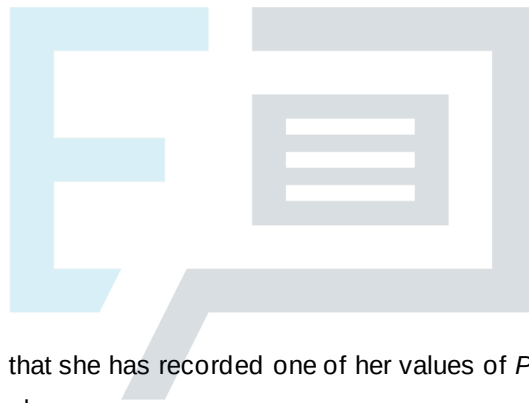
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A student conducts an experiment to investigate how P varies as h varies. She measures the value of P for various values of h , ensuring that all other conditions remain constant. Her results are summarised in the table below.

h (m)	0.40	2.50	3.60
P (s)	1.27	2.17	3.81

(b) Show that these results are not consistent with the answers to part (a).

[2]



(c) The student later realises that she has recorded one of her values of P incorrectly.

- Identify the incorrect value.
- Estimate the correct value that she should have recorded.

[3]

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- 11 A student is investigating fluid flowing through a pipe.

In her first model she assumes a relationship of the form $P = Sp^\alpha g^\beta h^\gamma$ where ρ is the density of the fluid, h is the length of the pipe, P is the pressure difference between the ends of the pipe, g is the acceleration due to gravity and S is a dimensionless constant. You are given that ρ is measured in kg m^{-3} .

- (a) Use the fact that pressure is force per unit area to show that $[P] = \text{ML}^{-1}\text{T}^{-2}$.

[2]

- (b) Find the values of α , β and γ .

[6]

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The density of the fluid the student is using is 540 kg m^{-3} . In her experiment she finds that when the length of the pipe is 1.40 m the pressure difference between the ends of the pipe is 3.25 N m^{-2} .

- (c) Find the length of the pipe for which her first model would predict a pressure difference between the ends of the pipe of 4.65 N m^{-2} .

[2]



In an alternative model the student suggests a modified relationship of the form $P = S\rho^\alpha g^\beta h^\gamma + \frac{1}{2}\rho v^2$, where v is the average velocity of the fluid in the pipe.

- (d) Use dimensional analysis to assess the validity of her alternative model.

[2]

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- 12 (a) Write down the dimension of density. [1]

The workings of an oil pump consist of a right, solid cylinder which is partially submerged in oil. The cylinder is free to oscillate along its central axis which is vertical. If the base area of the pump is 0.4 m^2 and the density of the oil is 920 kg m^{-3} then the period of oscillation of the pump is 0.7 s .

A student assumes that the period of oscillation of the pump is dependent only on the density of the oil, ρ , the acceleration due to gravity, g , and the surface area, A , of the circular base of the pump. The student attempts to test this assumption by stating that the period of oscillation, T , is given by $T = C \rho^\alpha g^\beta A^\gamma$ where C is a dimensionless constant.

- (b) Use dimensional analysis to find the values of α , β and γ . [4]

- (c) Hence give the value of C to 3 significant figures. [2]

- (d) Comment, with justification, on the assumption made by the student that the formula for the period of oscillation of the pump was dependent on only ρ , g and A . [2]

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The universal law of gravitation states that $F = \frac{Gm_1m_2}{r^2}$ where F is the magnitude of the force between two objects of masses m_1 and m_2 which are a distance r apart and G is a constant.

Find the dimensions of G .

[4]



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